

Week	Lesson	Contents	Stewart's book sections
1	1	Numbers, inequalities and absolute values. Coordinate geometry and lines.	Appendixes A and B
	2	Graphs of second-degree equations. Trigonometry.	Appendixes C and D
2	3	Four ways to represent a function. Mathematical models: A catalog of essential functions.	Sections 1.1-1.2
	4	New functions from old functions. Exponential functions. Inverse functions and logarithms.	Sections 1.3-1.5
3	5	The tangent and velocity problems. The limit of a function. Calculating limits using the limit laws.	Sections 2.1-2.3
	6	The precise definition of a limit.	Section 2.4
4	7	Continuity. Limits at infinity. Horizontal asymptotes.	Sections 2.5-2.6
	8	Derivatives and rates of change. The derivative as a function.	Sections 2.7-2.8
5	9	Derivatives of polynomials and exponential functions. The product and quotient rules. Derivatives of trigonometric functions.	Sections 3.1-3.3
	10	The chain rule. Implicit differentiation. Derivatives of logarithmic functions. Hyperbolic functions.	Sections 3.4-3.6, 3.11
6	11	Maximum and minimum values. The mean value theorem.	Sections 4.1-4.2
	12	How derivatives affect the shape of a graph. Indeterminate forms and L'Hôpital's rule.	Sections 4.3-4.4
7	13	Summary of curve sketching. Optimization problems. Antiderivatives	Sections 4.5, 4.7, 4.9
	14	Areas and distances. The definite integral.	Section 5.1-5.2
8	15	The fundamental theorem of calculus. Indefinite integrals and the net change theorem. The substitution rule.	Sections 5.3-5.5
	16	Areas between curves. Volumes. Average value of a function.	Sections 6.1-6.2, 6.5
9	17	Integration by parts. Trigonometric integrals. Trigonometric substitution.	Sections 7.1-7.3
	18	Integration of rational functions by partial fractions.	Section 7.4
10	19	Strategy for integration. Improper integrals.	Sections 7.5, 7.8
	20	Arc length. Area of a surface of revolution.	Sections 8.1-8.2
11	21	Sequences.	Section 11.1
	22	Series. The integral test and estimates of sums. The comparison tests.	Sections 11.2-11.4
12	23	Alternating series. Absolute convergence and the ratio and root tests. Strategy for testing series.	Sections 11.5-11.7
	24	Power series. Representations of functions as power series.	Sections 11.8-11.9
13	25	Taylor and Maclaurin series.	Sections 11.10
	26	Summarizing	Exercises